Octahedral dimer/trimer coverings

First exact classical calculation

Model used

Corner-sharing octahedra with cubic centres. Covering sites lie at cubic-bond midpoints. Allow any subset of the three diameter dimers or one face trimer; forbid two complementary trimers.
Every complete allowed covering is assigned equal statistical weight.

Feasibility demonstrated

Computed periodic sizes 3, 4, 6, 8, 10, 12 cubed.
Largest: 1,728 octahedra and 5,184 covering sites.
The binary constraint solve took 0.034 seconds.
It represents 2^1763 coverings without enumerating them.

Exact local results

Each diameter dimer has probability 1/4. Any trimer: 1/2.
Each specified triangular-face trimer: 1/16.
Connected correlations of single-octahedron observables vanish beyond neighboring centres. There is no nonzero decay tail to fit.

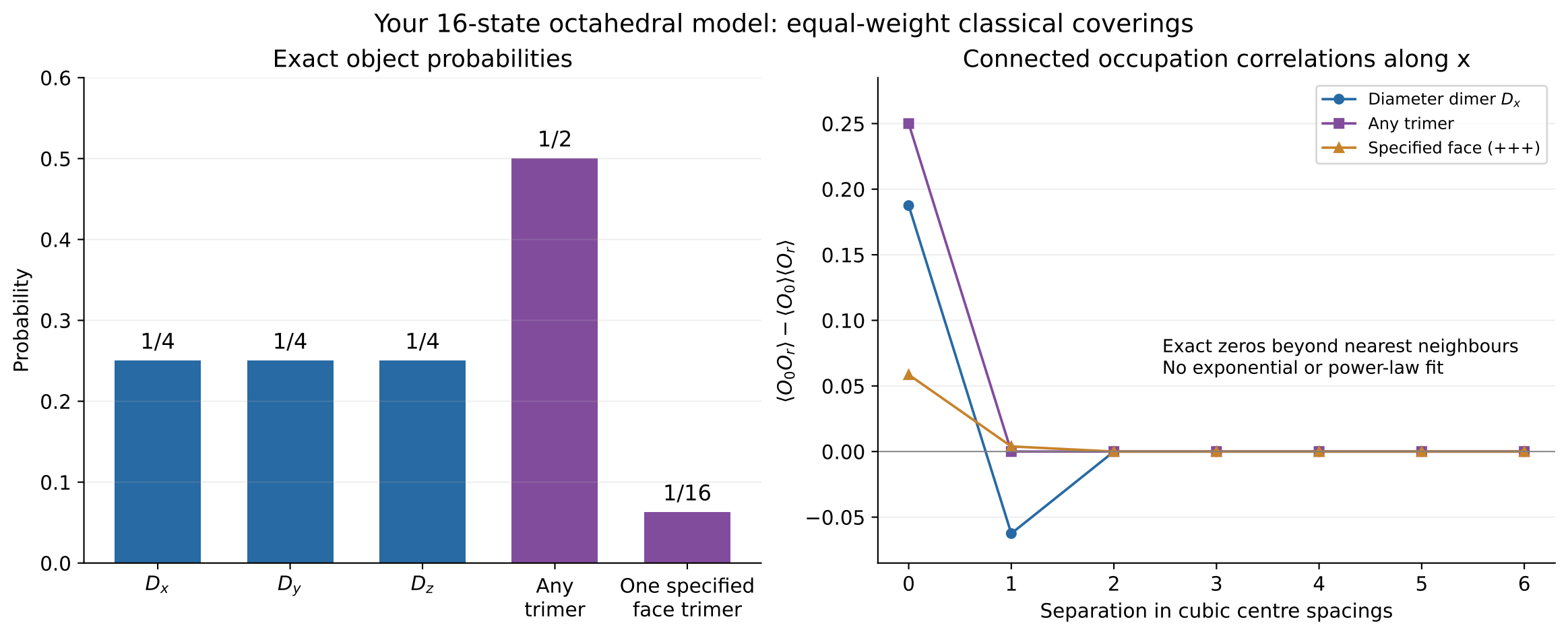
Nonlocal structure remains

For N=Lx Ly Lz and S=Lx+Ly+Lz, the covering count is 2^(N+S-1).
Cube-only moves leave 2^(2S-3) disconnected classical sectors.
The trimer parity on every periodic coordinate line is fixed by the line length. Zero two-point tails do not remove these constraints.

Scope

This is a classical covering calculation. No quantum dimer/trimer state or Hamiltonian was assumed. Nonuniform weights would require new calculations. The cited ten-state Ising model has different constraints and its correlation results do not transfer.

Reproduce: python scripts/run_octahedral_classical.py
Raw results and a checked sample covering accompany this PDF.


Your 16-state octahedral model: equal-weight classical coverings
Exact object probabilities
Probability
0.0
0.1
0.2
0.3
0.4
0.5
0.6
1/4
1/4
1/4
1/2
1/16
D_x
D_y
D_z
Any trimer
One specified face trimer
Connected occupation correlations along x
$\langle O_0 O_r \rangle - \langle O_0 \rangle \langle O_r \rangle$
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
Diameter dimer D_x
Any trimer
Specified face (+++)
Exact zeros beyond nearest neighbours
No exponential or power-law fit
0
1
2
3
4
5
6
Separation in cubic centre spacings